

Jose Bande Serrano

Full-stack Software Engineer

New York, NY | bandeserrano@gmail.com | (347) 918-6512 | linkedin.com/in/bandeserrano

Professional Summary

Senior Full-stack Software Engineer with 10+ years of experience building production web applications across frontend, backend, cloud, and workflow automation systems, including recent hands-on delivery of AI-powered applications. Offers strong evidence in OpenAI integration, AI agents, RAG-style retrieval workflows, document-processing systems, React, Next.js, Node.js, Python, REST APIs, and Azure-based application delivery. Background includes scalable system architecture, cloud-native services, complex UI development, and production automation platforms used in real business operations.

Selected Skills

OpenAI, AI Agents, RAG, MCP Server, React, Next.js, Node.js, Python, REST APIs, Azure, Azure Functions, System Architecture, Workflow Orchestration, Document Processing, TypeScript, CI/CD

Professional Experience

NBCUniversal - Full-Stack Engineer, Compliance Automation

2024-09 - present | New York, NY

Built and delivered a production compliance automation platform combining a high-performance Next.js SPA, backend workflows, ETLs, and reporting systems.

- Implemented AI bots integrated with Microsoft Teams.
- Designed and implemented tens of Dataverse tables to support the application's data model.
- Created dozens of Power Automate flows and dataflows to support compliance workflows and data movement.
- Built a React.js and Next.js application that is in production and used by hundreds of NBCUniversal employees.
- Developed a React application embedded inside a model-driven Power Apps application.

Enerex - AI Development Consultant

2023-11 - present | Remote - New Jersey

Architected AI-powered automation platforms spanning user-facing applications, asynchronous workflow orchestration, and cloud delivery for contract-processing systems.

- Architected the "Unattended Mode" platform, an asynchronous work-farm architecture for AI-driven email and contract processing without requiring users to wait in front of a UI.
- Designed the platform around three core services: a connector API for task intake and third-party email ingestion, a monitor application for task and flow persistence plus observability, and a queue-triggered Azure Functions worker tier for agent execution.
- Implemented agents with RAG via tool functions and an MCP server so workflows could retrieve grounded business context during contract-processing tasks.
- Created abstractions that made it easier for developers to add new agents and compose new AI workflows on top of the platform.

- Provisioned and deployed Azure Functions, web applications, databases, and Storage Queues to support Enerex products.
- Implemented a queue-driven execution model using Azure Storage Queues and Azure Functions so tasks could be processed in parallel and the platform could scale horizontally under background workload demand.
- Built an AI-powered Outlook add-in to automate contract processing workflows for energy retail market users.

Broadcasting Platforms - Senior Frontend Engineer & Team Lead

2019-11 - 2022-12 | Remote

Led and built the core user-facing frontend applications of a real-time streaming platform, with deep ownership across Vue.js architecture, design-system implementation, and WebSocket-driven interaction systems.

- Improved frontend delivery performance by introducing HTTP/2, asset compression, and image conversion to WebP format.
- Built two core user-facing streaming applications in Vue.js that formed the frontend backbone of the platform.
- Designed and implemented an SEO-focused rendering solution that served server-side-rendered content to web crawlers while preserving the SPA experience for regular users.
- Implemented demanding frontend experiences from professional Figma design specifications with strong attention to design fidelity and usability.
- Used Storybook as a design-system layer to centralize, standardize, and control component styling across the platform.

First Due - Senior Software Engineer

2017-07 - 2019-11 | Remote

Built large-scale web crawlers and ETL pipelines with Python and event-driven backend systems.

- Designed and maintained PostgreSQL databases supporting large-scale data collection workloads.
- Implemented tens of Python web crawlers with the Scrapy framework to extract critical company data from public websites.
- Architected a system to manage and control web crawlers, centralizing execution control and operational monitoring in a single UI.
- Built crawler orchestration and monitoring capabilities using Django, FastAPI, and RabbitMQ.
- Improved control and accountability of company web-crawling operations through centralized management and visibility.

Liberty Power - .NET/SQL Consultant

2016-05 - 2017-06 | Remote

Modernized legacy .NET applications and improved database quality and performance with SQL work.

- Implemented and ran SQL queries to investigate, validate, and repair data issues.
- Migrated projects from Visual Basic to ASP.NET as part of legacy application modernization work.
- Implemented numerous stored procedures to support application behavior and database operations.
- Performed extensive database data consistency reviews and corrections across business-critical datasets.

Projects

AI Assisted Invoice Processor

An application to help finance teams collect invoices, extract structured data, review exceptions, and move approved records into downstream reporting without juggling disconnected inboxes, sheets, and shared folders.

- Implemented AI-powered invoice analysis using OpenAI.
- Used Azure Functions to run agents that analyze incoming invoices and return structured results for downstream processing.
- Built the frontend with React.js, Next.js, Tailwind CSS, Lucide React, and TanStack to support real-time, data-driven invoice operations.
- Integrated Nylas and Google services to enable controlled access to user email inboxes for invoice ingestion.
- Built the backend with Node.js, Express.js, and Prisma over SQL Server hosted in Azure.

Education

Bachelor's Degree, Engineer in Electronics

Ciudad Universitaria Jose A. Echeverria (CUJAE) | Havana, Cuba | 2004 - 2009